

Aditya Kaushikbhai Rathod

B.Tech. in Electrical Engineering | Indian Institute of Technology, Jodhpur

+91-9726608281 | aditya.rathod2912@gmail.com | GitHub: [aditya8281](https://github.com/aditya8281) | Website: aditya8281.github.io | LinkedIn: [Aditya Rathod](#)

EDUCATION

Indian Institute of Technology, Jodhpur

2024 – Present

B.Tech. in Electrical Engineering | CGPA: 7.38/10

Relevant Coursework: Data Structures & Algorithms, Digital Design, Pattern Recognition & Machine Learning, Probability, Statistics & Stochastic Processes, Mathematics I & II, Introduction to Computer Science, Foundation of Quantum Information

PROFESSIONAL SUMMARY

- Software engineering candidate focused on backend systems, Rust/Python development, and AI developer tooling; built CORTEX with **180+ FastAPI endpoints**, **10 domain routers**, and **1,900+ automated tests**.
- Built ASTERA, a multi-language code-intelligence engine supporting **14 programming languages** for repository-scale static analysis, semantic search, and impact reasoning; validated with **153 automated tests** and **92 Criterion benchmarks**.
- Evaluated **30+ CNN-Attention architectures** in PyTorch using controlled ablations on imbalanced PlantVillage and Mendeley Potato Leaf Disease datasets.

TECHNICAL SKILLS

Languages: Python, C, C++, Rust, TypeScript, JavaScript, SQL, Embedded C/C++

Systems: Data Structures, Algorithms, Object-Oriented Programming, System Design, Operating Systems, Computer Networks, Concurrency, Multithreading, Synchronization, Async Programming, Memory Management, IPC, Tokio, Rayon

Backend & Infrastructure: FastAPI, Axum, REST APIs, WebSockets, Microservices, Pydantic, SQLAlchemy, Alembic, Redis, Linux, Docker, GitHub Actions, Docker Compose, CI/CD, API Design

Databases & Search: PostgreSQL, SQLite, SQLite FTS5, Qdrant, pgvector, Vector Search, Semantic Search, Full-Text Search, BM25, HNSW

AI & Developer Tools: PyTorch, Machine Learning, OpenCV, YOLO, Tree-sitter, Static Analysis, Code Property Graphs, Program Analysis, RAG, Embeddings, Agent Systems, Ollama

Engineering Tools: Git, pytest, Unit Testing, Integration Testing, Debugging, Code Review, Serde, Clap, Notify, Tracing, Authentication, Rate Limiting

EXPERIENCE

•Undergraduate Researcher – Computer Vision, IIT Jodhpur

| May 2026 – Present | [GitHub](#)

Robust Classification for Imbalanced Natural Image Datasets

PyTorch, Computer Vision, Reproducible ML Evaluation

- Reproduced PLDNet as the baseline and fixed dataset splits, training settings, and evaluation scripts before comparing CNN-Attention variants on imbalanced image data.
- Evaluated **30+ model variants** with ablations and feature-fusion changes to compare robustness under class imbalance and real-world image variation.
- Analyzed results on Mendeley Potato Leaf Disease and PlantVillage datasets, comparing class-imbalance behavior instead of relying only on headline accuracy.

•Open Source Contributor – plink, Devlup Labs IIT Jodhpur

| May 2025 – Nov. 2025 | [GitHub](#)

Secure Peer-to-Peer File Transfer across Heterogeneous Networks

Python, TCP/IP, Sockets, Multithreading, AES-256, NAT Traversal, Git

- Contributed TCP socket paths for encrypted file chunks, metadata exchange, and multithreaded transfer across heterogeneous networks.
- Completed **16 assigned tasks** across cryptography, networking research, file metadata, chunk handling, parsing refactors, port scanning, UDP hole-punching prototypes, direct-connection management, network-type detection, and documentation.

ENGINEERING PROJECTS

•ASTERA

| Mar. 2026 – Present | [GitHub](#)

Multi-Language Static Analysis & Code Intelligence Engine

Rust, Tree-sitter, SQLite FTS5, Axum, Tokio, Rayon, REST APIs, Program Analysis

- Parsed repositories across **14 programming languages** into a queryable **Code Property Graph** for symbol lookup, dependency traversal, impact analysis, broken-reference detection, and semantic search.
- Built a **12-crate Rust pipeline** for file discovery, Tree-sitter parsing, symbol extraction, graph-edge construction, SQLite FTS5 indexing, REST APIs, and exports.
- Covered parser, extractor, storage, REST API, search, graph traversal, and metrics paths with **153 automated tests** and **92 Criterion benchmarks**.
- Optimized hot paths to **33 ns** language classification, **16.16 µs** SQLite FTS5 symbol search, **3–24 µs** impact traversals, and **229 ms** metrics over 10,000 graph nodes.

•CORTEX

| Jan. 2026 – Present | [GitHub](#)

Local-First Machine Intelligence Layer

Python, FastAPI, Next.js, PostgreSQL, Redis, Qdrant, JWT, Agent Systems

- Designed **hybrid retrieval** that combines semantic memory, episodic memory, working memory, vector search, and knowledge-graph recall over repositories, documents, conversations, and project history.
- Implemented asynchronous services for repository analysis, file tracking, GitHub sync, memory search/decay, model catalogs, providers, and recommendations across a **180+ endpoint FastAPI backend**.
- Implemented the **agent run loop** for tool execution, output verification, stall detection, context compaction, token accounting, event handling, and security policy checks.
- Validated agents, REST APIs, memory, privacy, services, and infrastructure with **1,900+ automated tests** across **10 domain routers**.

•Parallel Blockchain Simulator

| Nov. 2025 – Dec. 2025 | [GitHub](#)

Concurrent Blockchain Simulation with Thread-Safe Validation

Python, Multithreading, SHA-256, Queues, Hash Maps, Merkle Trees, Mutex Locks

- Coordinated **7 blockchain threads**—3 miners, 3 validators, and 1 transaction generator—with mutex locks and events around shared chain state.
- Modeled balances, linked-list blocks, queued transactions, hash-map mempool lookup, Merkle verification, and SHA-256 Proof-of-Work at difficulty 3.
- Stress-tested synchronization under randomized transaction loads, observing **131–200 processed transactions** and **20–59 mined blocks** per run without corrupting the chain.

LEADERSHIP & CAMPUS INVOLVEMENT

Secretary, Designers – The Design Club, IIT Jodhpur

Aug. 2025 – Apr. 2026

• Guided **50+ students**, ran design workshops, and secured **7th rank** at Inter IIT Cultural Meet 8.0.

Student Guide, Student Wellbeing Committee (SWC), IIT Jodhpur

Aug. 2025 – Apr. 2026

• Supported **12 first-year students** through academic transition and campus onboarding.

Core Member, Inside – The Game Development Society, IIT Jodhpur

Aug. 2025 – Apr. 2026

• Coached **5 first-year students** on an Inside-backed game-building project.

Assistant Head, Prometeo'26 – IIT Jodhpur

Oct. 2025 – Jan. 2026

• Led **30+ volunteers** and managed digital design for Prometeo'26.